

Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

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Structure-preserving representation before decision contraction.

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pip install erisml-compiler · github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc

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Don't collapse to a scalar.

	harm	rights	consent	risk	fair
speaker	red	orange	purple	orange	red
village	red	red	orange	red	orange
refugees	green	green	green	orange	green
nazis	orange	purple	orange	orange	purple

the moral tensor (who x what is owed)

✓ keep the structure

who is affected · what is owed
who consented · who bears imposed risk

contract to a decision only at the end,
and log the contraction:
weights · who lost · what residue remains

★ 0.42

one scalar — structure lost

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Four lenses. No silent winner.

one MoralGraph

Consequentialist
harm/care tensor
Gini · worst-off · Shapley

Deontic (Kantian)
gates · universalizability
via Z3 SMT

Virtue
Aristotelian
habit-consistency

Care ethics
Gilligan · Noddings
· Tronto

verdicts disagree? → cross_projection_disagreement

defers to a human

"We run four and show you where they conflict — we don't pick one and hide it."

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Safety in the loop. Fails safe.

Agent planner

Reflex
< 100 μs · hard stops

Tactical
ErisML · 10-100 ms

Strategic
policy + human oversight

ALLOW · REVISE · BLOCK

proof → proof → proof → proof

SHA-256 DecisionProof chain · previous_proof_hash → proof_hash

ethics service times out → rule-based fallback. Never fails open.

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pip install erisml-compiler — structure before the scalar

text

segment

extract

canonicalize

MoralGraph

tensorize

DEME

DecisionProof

MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

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Is this 'good'? → Does it preserve the bonds?

Obligation

Claim

Liberty

No-claim

O

C

L

N

s

r²

s = correlative swap
(agent ↔ patient)

r² = deontic negation

0.0 perfect

0.155 baseline

0.25 warn

0.30 block

Bond Index: does the judgment survive the swap?

multi-rank tensors: party x time x action x coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

s and r² commute → V4 (Klein four-group, order 4) — measured

full D4 (order 8) posited — quarter-turns not yet observed

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What the model says vs what it exhibits

transformer
(Qwen2.5-7B,
LLaMA, Mistral)

text lens
(what it says)

activation lens
(what it exhibits)

delta lens
compare

requires human review
(5 failure modes)

activation / probe lens is EARLY — uncalibrated by default (research-grade)

turboquant-pro (PyPi): PyTorch-native compression
HF 3-liner: generate(past_key_values=TurboQuantCache()); Triton + Volta kernels · vLLM · 5× KV
live in this stack: 3-bit embedding codec on the NATS memory bus · claims C1-gated (CLAIMS.md)

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Auditable means you can replay the judgment

stakeholder	harm	verdict
speaker	0.76	forbid
village	0.83	forbid
refugees	0.00	prefer
nazis	0.18	neutral

Gini(harm) = 0.43 · worst-off = village

Shapley: speaker 7.11 · refugees 7.70
nazis 7.88 · village 7.18

DecisionProof: proof_hash → audit.ir_hash

worked example examples/nazi_attic · rank-2 DEME — one command, real numbers, a hash you can verify.

One command. Real numbers.
A hash you can verify.

\$ erisml-compile examples/nazi_attic

per-party verdicts · Gini 0.43 · exact Shapley
DecisionProof chained to the IR hash — replay the judgment

- Structure before contraction — keep the tensor, log the collapse + the residue.
- Pluralism is the responsible move — four lenses, defer to a human on conflict.
- Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early / uncalibrated
ranks 1–3 real, higher partial · V4 measured, D4 posited (Lean + Mathlib verified)
embodied = design target · conversational agents = live